

AMS 617 cat

R Markdown

a. Generate Appropriate Graphs to Present the Data

```
# Load necessary Libraries
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
## filter, lag
```

```
## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union
```

```
library(readr)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(tinytex)
```

```
# Load dataset
df <- read_csv("C:\\Users\\franc\\rdatascience\\projects\\Housing.csv", show_col_types = FALSE)

# Convert categorical variables to factors
categorical_columns <- c("mainroad", "guestroom", "basement",
                        "hotwaterheating", "airconditioning", "prefarea", "furnishingstatus")

df[categorical_columns] <- lapply(df[categorical_columns], as.factor)

# View the first few rows
head(df)
```

price	a...	bedroo...	bathrooms	stories	mainroad	guestroom	basem...	hotwaterheating
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<fct>	<fct>	<fct>
13.300	7.42	4	2	3	yes	no	no	no

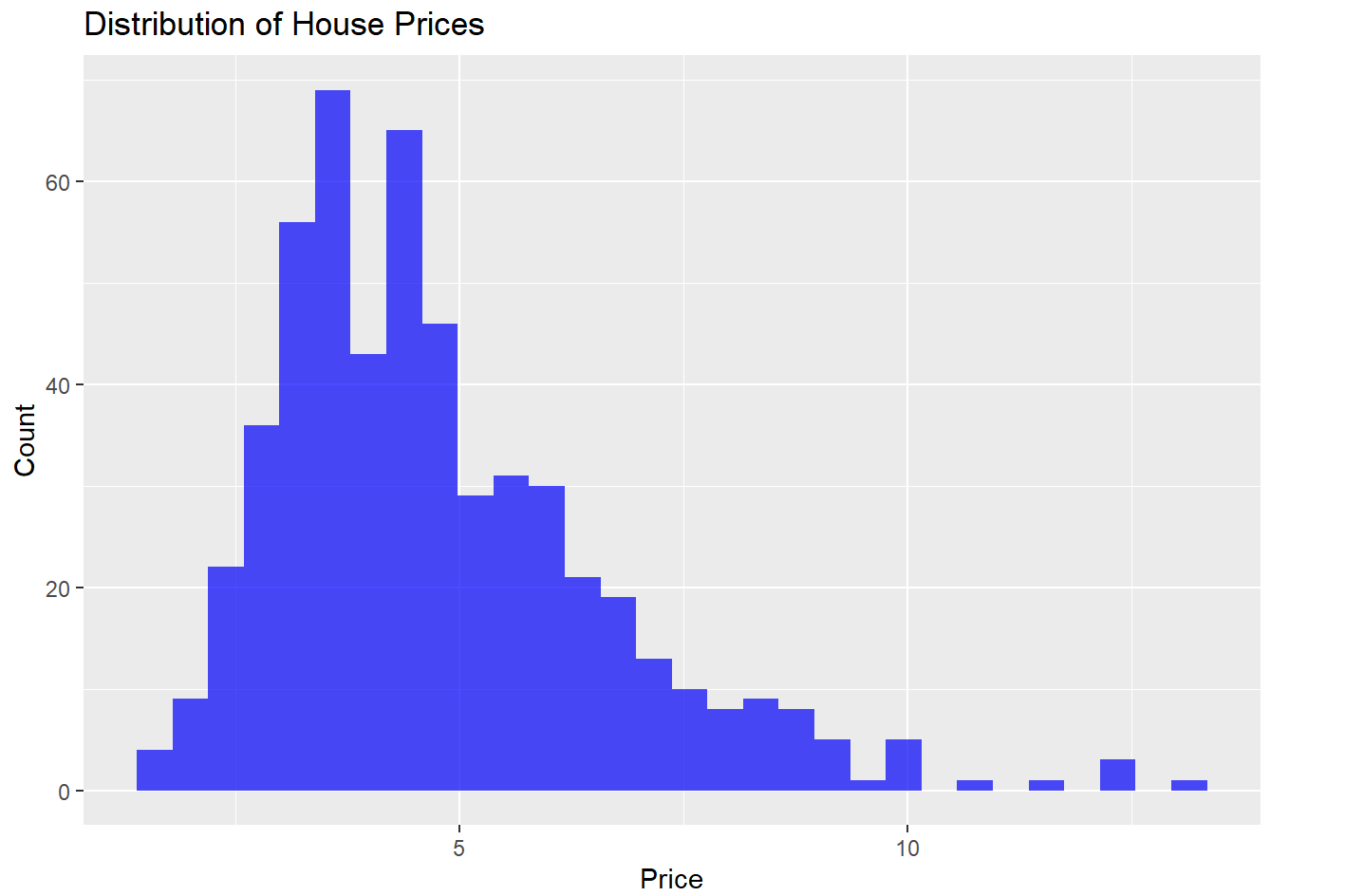
price	a...	bedroo...	bathrooms	stories	mainroad	guestroom	basem...	hotwaterheating	
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<fct>	<fct>	<fct>	
12.250	8.96	4	4	4	yes	no	no	no	
12.250	9.96	3	2	2	yes	no	yes	no	
12.215	7.50	4	2	2	yes	no	yes	no	
11.410	7.42	4	1	2	yes	yes	yes	no	
10.850	7.50	3	3	1	yes	no	yes	no	

6 rows | 1-9 of 13 columns

Explanation show_col_types = FALSE → Suppresses the message. as.factor() → Converts categorical (chr) columns to factor. lapply(df[categorical_columns], as.factor) → Converts all specified categorical columns.

Histogram of House Prices

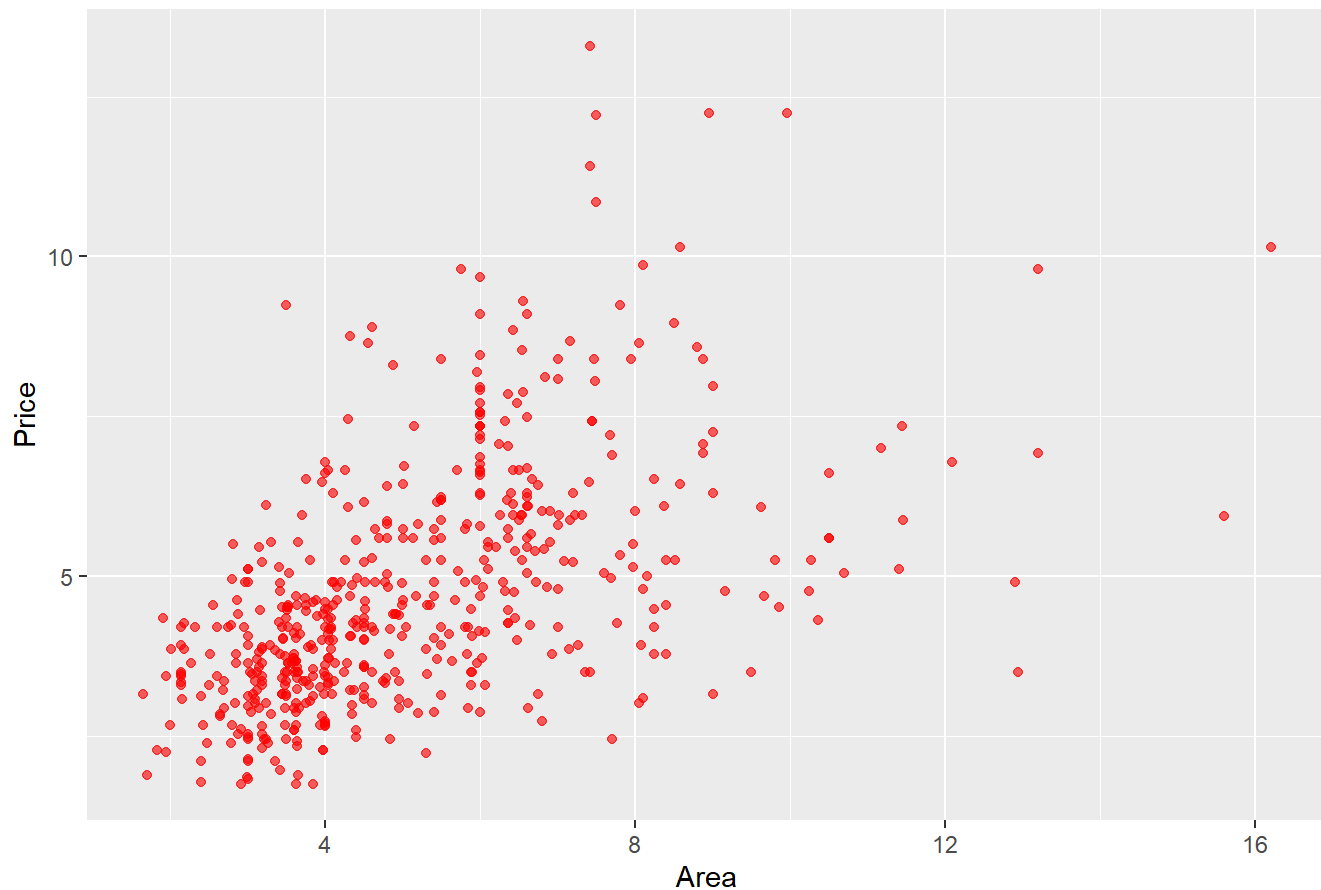
```
ggplot(df, aes(x = price)) +  
  geom_histogram(fill = "blue", bins = 30, alpha = 0.7) +  
  labs(title = "Distribution of House Prices", x = "Price", y = "Count")
```



Scatter Plot: Price vs. Area

```
ggplot(df, aes(x = area, y = price)) +
  geom_point(color = "red", alpha = 0.6) +
  labs(title = "Scatter Plot: Price vs Area", x = "Area", y = "Price")
```

Scatter Plot: Price vs Area

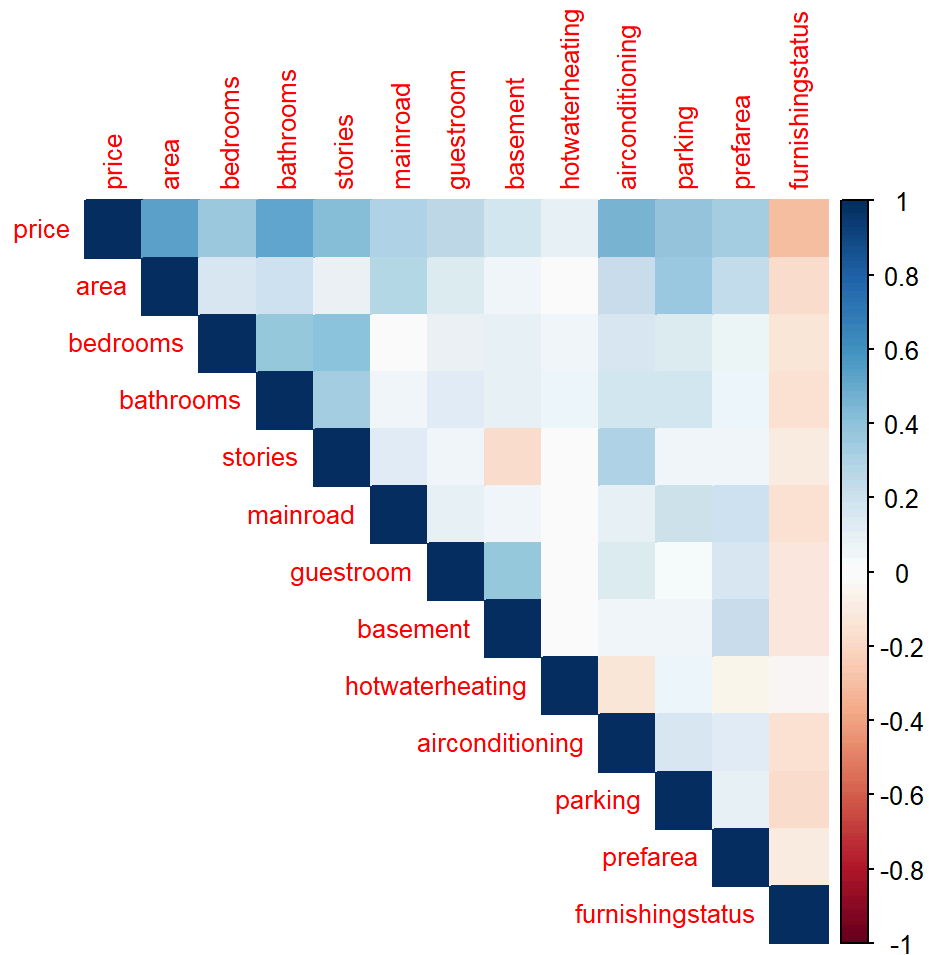


Correlation Heatmap

```
# Convert categorical variables to numeric
df_numeric <- df %>%
  mutate(across(where(is.character), as.factor)) %>%
  mutate(across(where(is.factor), as.numeric))

# Compute correlation matrix
corr_matrix <- cor(df_numeric)

# Heatmap
corrplot(corr_matrix, method = "color", type = "upper", tl.cex = 0.8)
```



`mutate(across(where(is.factor), as.numeric))` → Converts factors to numeric for correlation.

b. Perform Descriptive Analysis of the Dataset

```
# Summary statistics
summary(df)
```

```
##      price          area      bedrooms      bathrooms
## Min.   : 1.750    Min.   : 1.650    Min.   :1.000    Min.   :1.000
## 1st Qu.: 3.430    1st Qu.: 3.600    1st Qu.:2.000    1st Qu.:1.000
## Median : 4.340    Median : 4.600    Median :3.000    Median :1.000
## Mean   : 4.767    Mean   : 5.151    Mean   :2.965    Mean   :1.286
## 3rd Qu.: 5.740    3rd Qu.: 6.360    3rd Qu.:3.000    3rd Qu.:2.000
## Max.   :13.300    Max.   :16.200    Max.   :6.000    Max.   :4.000
##      stories    mainroad  guestroom  basement  hotwaterheating  airconditioning
## Min.   :1.000    no : 77    no :448    no :354    no :520          no :373
## 1st Qu.:1.000    yes:468    yes: 97    yes:191    yes: 25          yes:172
## Median :2.000
## Mean   :1.806
## 3rd Qu.:2.000
## Max.   :4.000
##      parking    prefarea    furnishingstatus
## Min.   :0.0000    no :417    furnished      :140
## 1st Qu.:0.0000    yes:128    semi-furnished:227
## Median :0.0000          unfurnished   :178
## Mean   :0.6936
## 3rd Qu.:1.0000
## Max.   :3.0000
```

c. Formulate Five Objectives, Perform Tests of Hypothesis, and Discuss Results

If $p\text{-value} \leq 0.05$, reject H_0 (significant effect).

If $p\text{-value} > 0.05$, fail to reject H_0 (no significant effect).

Objective 1: Does Area Affect Price?

Hypothesis:

H_0 : Area has no effect on price.

H_A : Area has an effect on price.

```
# Test: Pearson Correlation
cor_test <- cor.test(df$area, df$price)
cor_test
```

```
##
## Pearson's product-moment correlation
##
## data: df$area and df$price
## t = 14.795, df = 543, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.4733158 0.5932784
## sample estimates:
##      cor
## 0.5359973
```

conclusion: Area has an effect on price

Objective 2: Do Houses Near the Main Road Have Higher Prices?

Hypothesis:

HO: No price difference based on proximity to the main road.

HA: Houses near the main road have different prices.

```
# Test: t-test  
t.test(price ~ mainroad, data = df)
```

```
##  
## Welch Two Sample t-test  
##  
## data: price by mainroad  
## t = -11.853, df = 210.68, p-value < 2.2e-16  
## alternative hypothesis: true difference in means between group no and group yes is not equal  
## to 0  
## 95 percent confidence interval:  
## -1.857781 -1.327964  
## sample estimates:  
## mean in group no mean in group yes  
## 3.398905 4.991777
```

conclusion: Main Road has an effect on price.

p-value ≤ 0.05 , hence reject HO

Objective 3: Does the Number of Bedrooms Impact Price?

Hypothesis:

HO: Number of bedrooms does not affect price.

HA: Number of bedrooms affects price.

```
# Test: Linear Regression  
model_bedrooms <- lm(price ~ bedrooms, data = df)  
summary(model_bedrooms)
```

```
##
## Call:
## lm(formula = price ~ bedrooms, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.6967 -1.2103 -0.2653  0.9409  7.5721
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.0127     0.3092   6.510 1.71e-10 ***
## bedrooms      0.9288     0.1012   9.179 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.742 on 543 degrees of freedom
## Multiple R-squared:  0.1343, Adjusted R-squared:  0.1327
## F-statistic: 84.25 on 1 and 543 DF,  p-value: < 2.2e-16
```

conclusion: Number of Bedrooms has an effect on price since,

$p\text{-value} \leq 0.05$, hence reject H_0 (significant effect).

Objective 4: Is There a Significant Price Difference Between Furnished and Unfurnished Houses?

H_0 : No significant difference in price based on furnishing.

H_A : There is a significant difference.

```
# Test : ANOVA
anova_model <- aov(price ~ furnishingstatus, data = df)
summary(anova_model)
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## furnishingstatus  2  179.8   89.90   28.27 2.09e-12 ***
## Residuals       542 1723.4    3.18
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

conclusion: Level of finishing has an effect on price since, $p\text{-value} \leq 0.05$, hence reject H_0 .

Objective 5: Does Having Air Conditioning Increase the Price?

H_0 : Air conditioning has no effect on price.

H_A : Air-conditioned houses have higher prices.

```
# Test: t-test
t.test(price ~ airconditioning, data = df)
```

```
##  
##  Welch Two Sample t-test  
##  
## data:  price by airconditioning  
## t = -10.659, df = 262.47, p-value < 2.2e-16  
## alternative hypothesis: true difference in means between group no and group yes is not equal  
## to 0  
## 95 percent confidence interval:  
##  -2.157719 -1.484842  
## sample estimates:  
##  mean in group no mean in group yes  
##           4.191940           6.013221
```

conclusion: Having Air-conditioned houses has an effect on price since,
 $p\text{-value} \leq 0.05$, hence reject H_0 .